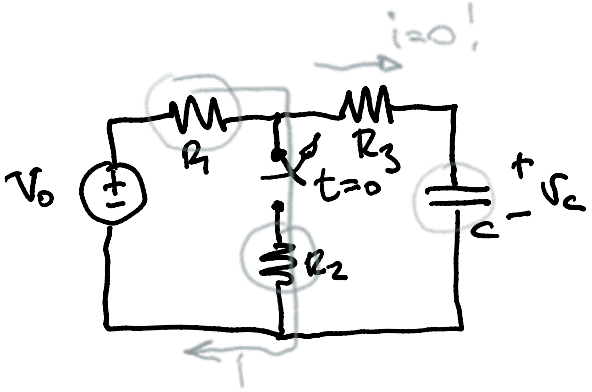
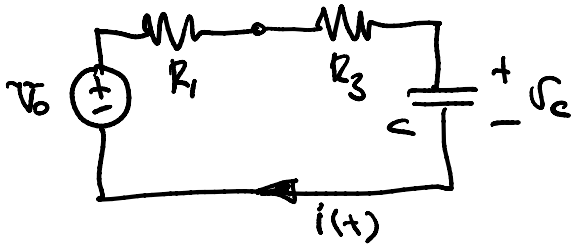
TENTAPROB 210819, 3. (VARIANT)



$V_0 = 80\ V$
$R_1 = 10\ k\Omega$
$R_2 = 3333\ \Omega$
$R_3 = 20\ k\Omega$
$C = 100\ nF$

STATIONÄRT FÖRHÅLLANDE FÖR $t < 0$.

LÖSNING: FÖR $t \geq 0$



KSR ger

$$-V_0 + (R_1+R_3)i + v_c = 0 \quad (1)$$

MEN VI VET ATT

$$v_c = v_c(0) + \frac{1}{C}\int i\,dt \quad (2)$$

EKV (2) I (1) ger

$$-V_0 + (R_1+R_3)i + v_c(0) + \frac{1}{C}\int i\,dt = 0 \quad (3)$$

DERIVERA MAP t

$$(R_1+R_3)\frac{di}{dt} + \frac{1}{C}i = 0$$

$$\frac{di}{dt} + \frac{1}{(R_1+R_3)C}i = 0$$

VI VET ATT $i(t) = Ae^{-\frac{1}{(R_1+R_3)C}t}$, $\frac{1}{(R_1+R_3)C} = ?$ (4)

FÖR ATT BESTÄMMA A MÅSTE VI STUDERA KRETSEN VID $t=0$

(3) ger för $t=0$

$$-V_0 + (R_1+R_3)i(0) + v_c(0) + \underbrace{\frac{1}{C}\int_0^0 i\,dt}_{=0} = 0$$

$$i(0) = \frac{V_0 - v_c(0)}{\quad} \quad (5)$$

$$i(0) = \frac{V_0 - V_c(0)}{R_1 + R_3} \qquad (5)$$

$\overrightarrow{=0}$

UR (4) για $t=0$

$$i(0) = A \overset{(5)}{=} \frac{V_0 - V_c(0)}{R_1 + R_3}$$

$$\Rightarrow \quad i(t) = \frac{V_0 - V_c(0)}{R_1 + R_3} e^{-\beta t} \qquad (6)$$

από (1)

$$-V_0 + (R_1 + R_3) i = -V_c$$

$$V_c(t) = V_0 - (R_1 + R_3) \frac{V_0 - V_c(0)}{R_1 + R_3} e^{-\beta t} =$$

$$= V_0 - (V_0 - V_c(0)) e^{-\beta t}$$

$$= V_0 (1 - e^{-\beta t}) + V_c(0) e^{-\beta t}$$

